

Gradient descent

Setup

- Set seed with value 42
- Set default datatype for Pytorch as double

Data loading

The `load_cifar` function loads the CIFAR10 data while only keeping the labels airplane and bird. We use the default train and validation split 90% training data and 10% validation data. Then we are left with the following sizes of each set:

- Train size: 8980
- Validation size: 1020
- Test size: 2000

Preprocessing and analysis

Firstly we normalize the data using `Pytorch.transforms.Normalize` with the mean and standard deviation of all the images that we have in the training set.

The class distribution was almost 50/50 with around 49.8% of the images in the training set being birds, and the rest being planes. We also map the labels to binary so that we have `plane=0` and `bird=1`.

The effect of normalization on model performance is illustrated in Figure 1.

